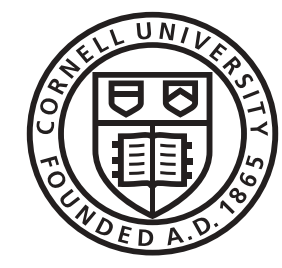


Shifting Mechanisms:

How Positional Encoding Choice Shapes In-Context Retrieval

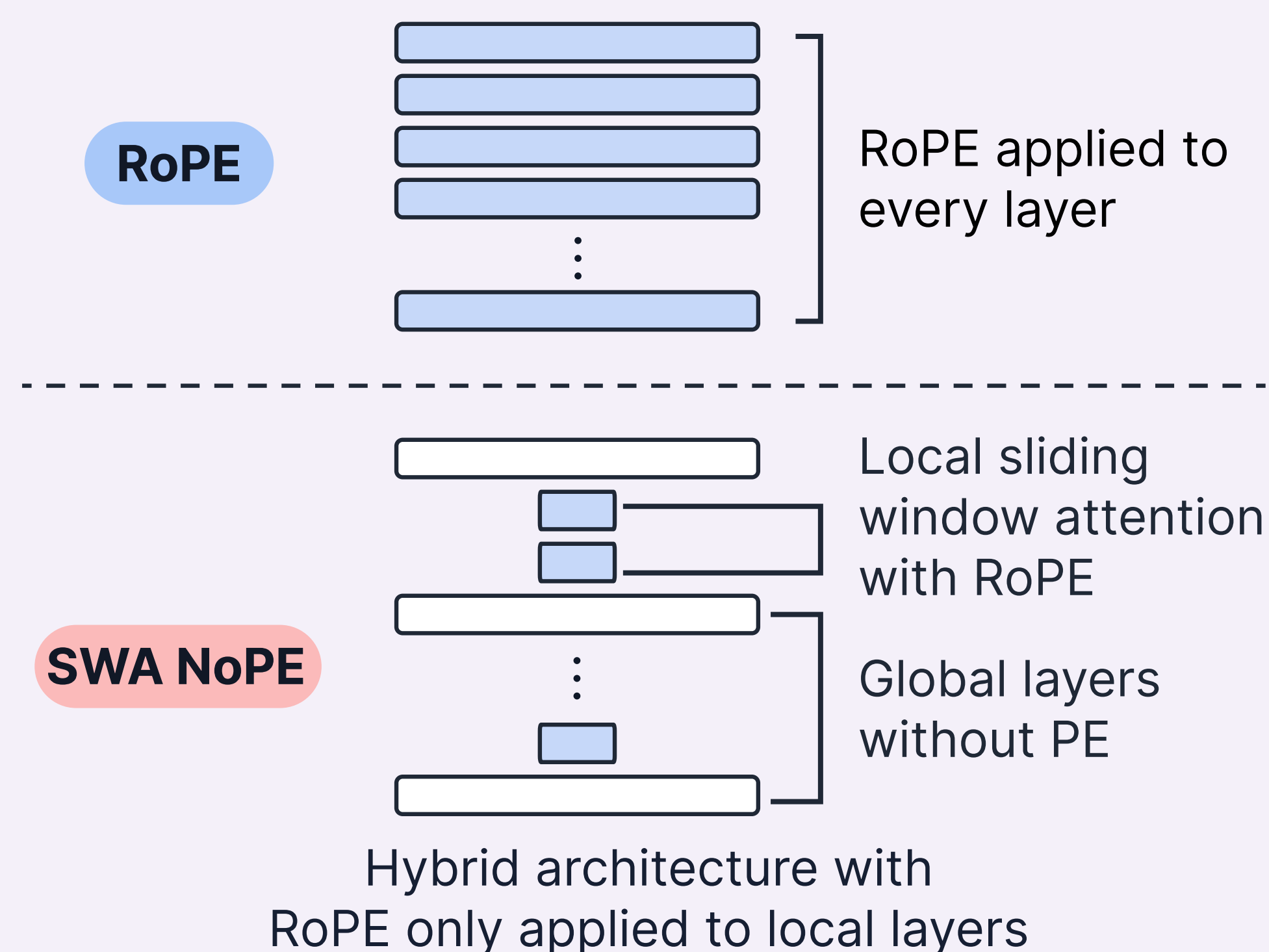
Eric Enouen, Sainyam Galhotra



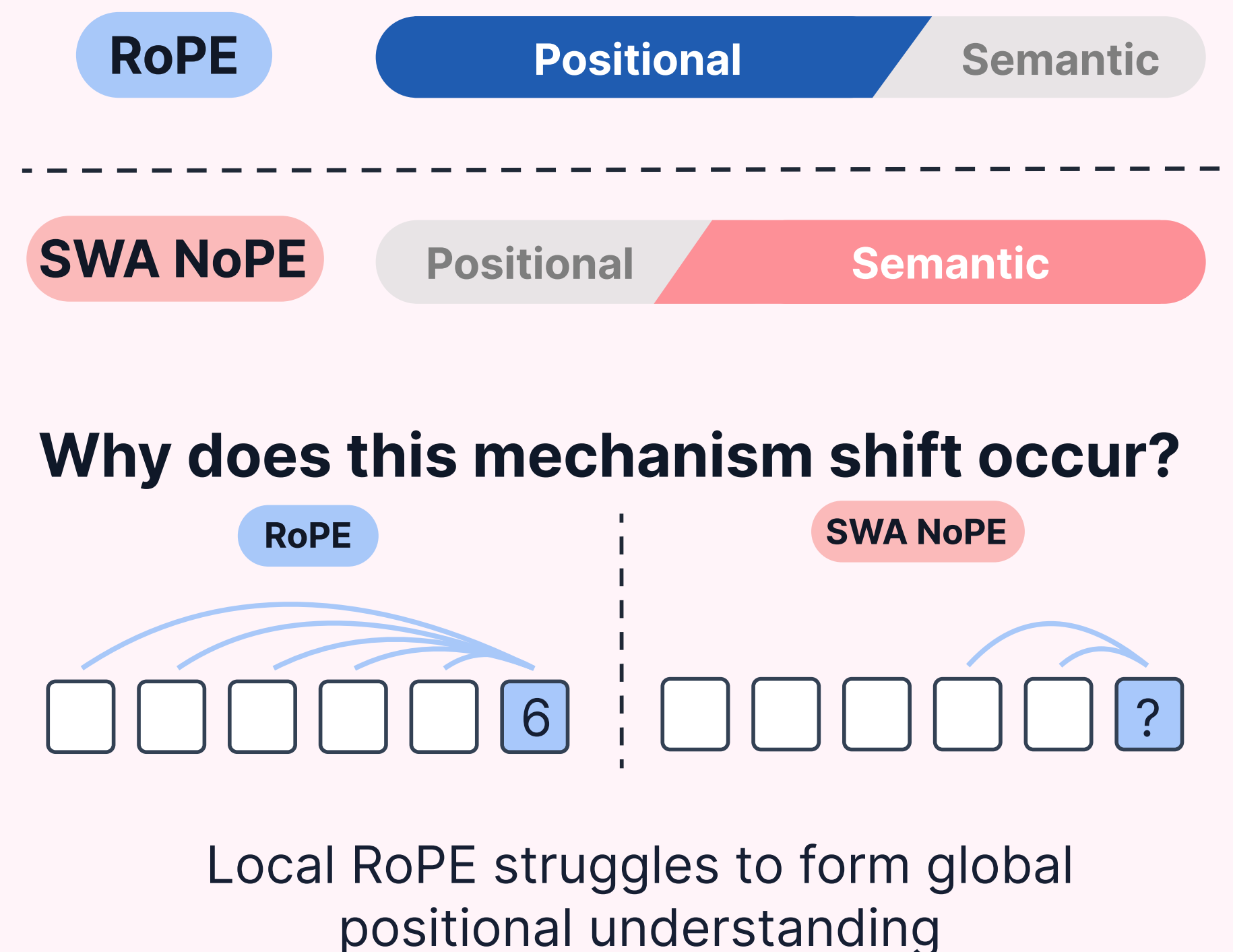
Cornell University



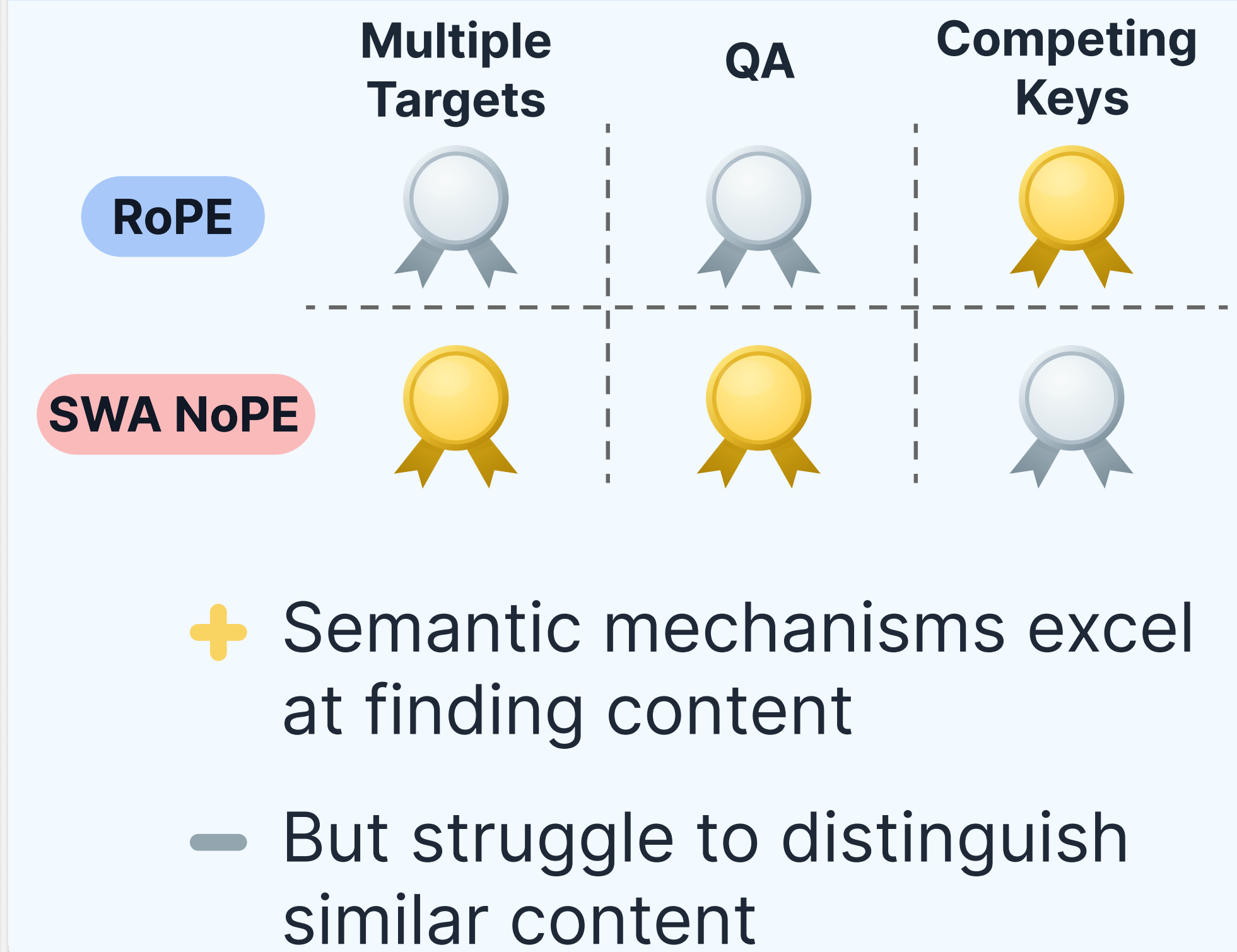
Positional Encoding Choice



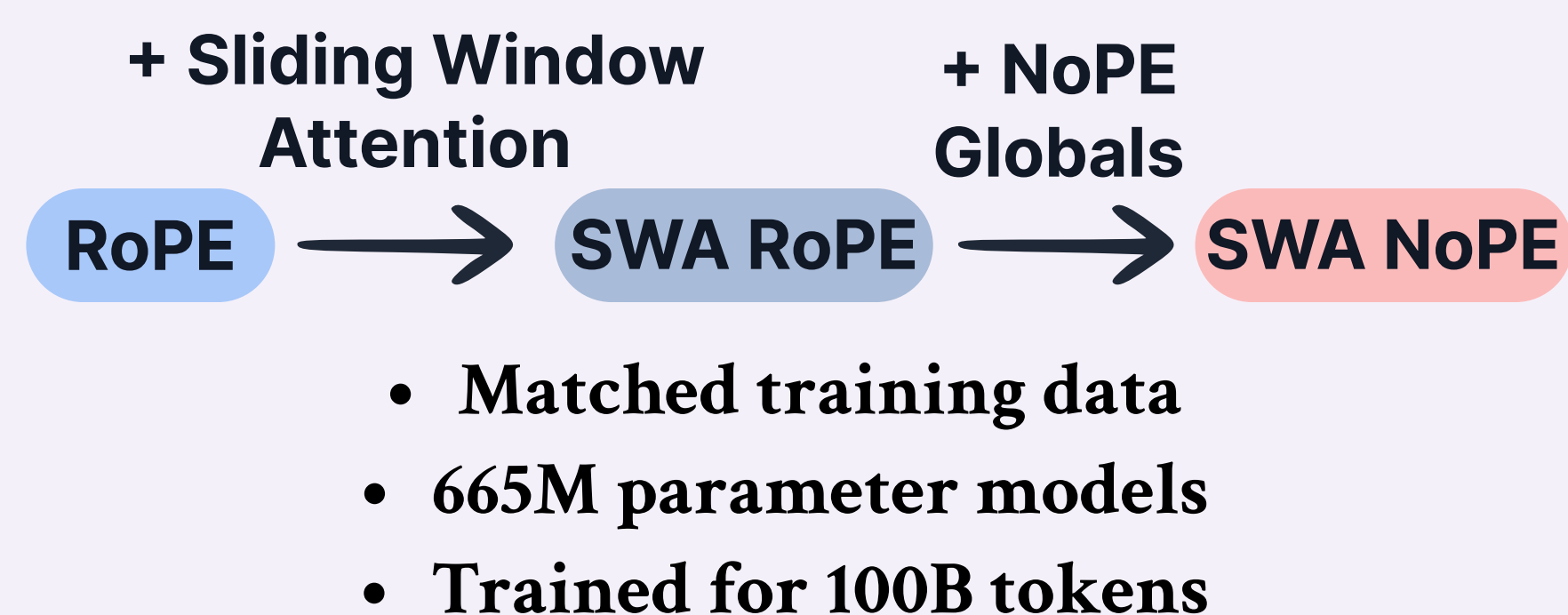
Mechanism Allocation



Long-Context Retrieval



Controlled Pre-training Ablation



Confining PE Drives the Shift

MECHANISM ALLOCATION	
RoPE	0.26
SWA RoPE	0.27
SWA NoPE	0.13

SWA NoPE shifts allocation compared to RoPE/SWA RoPE

Model Ranking Reverses

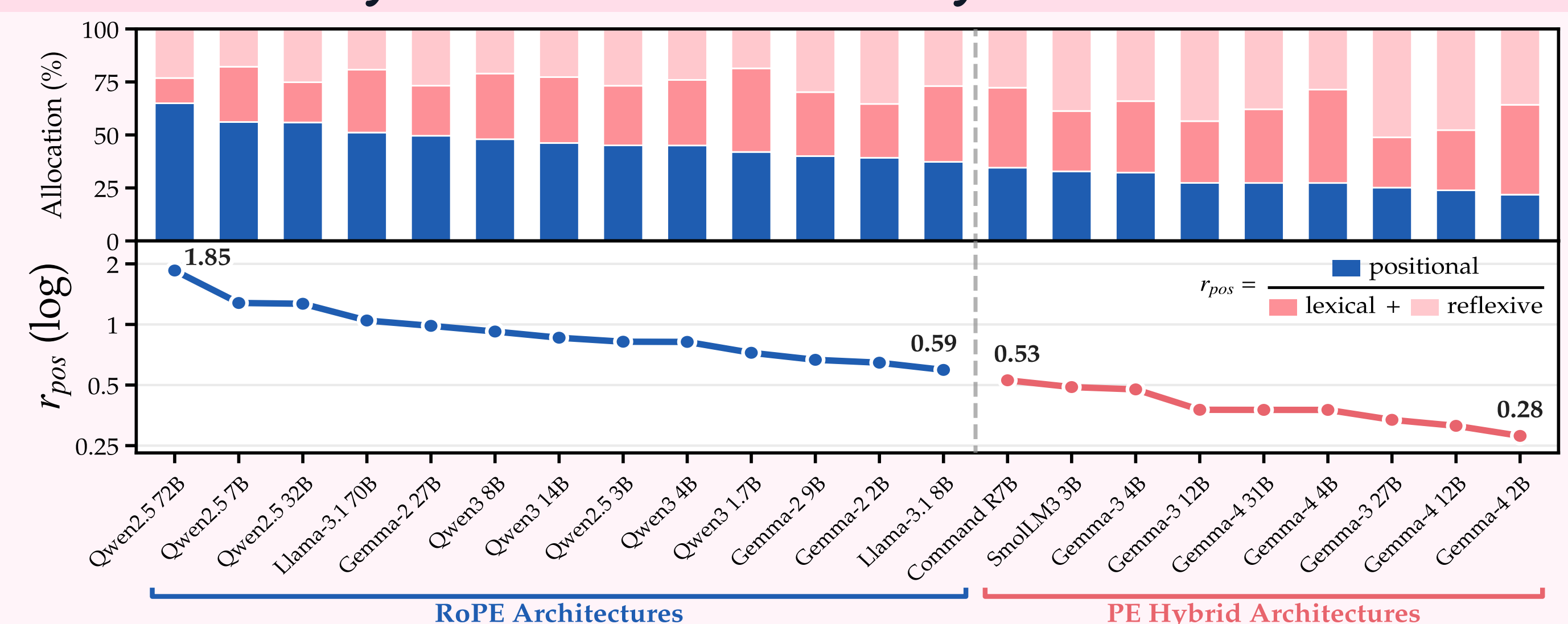
MULTI-QUERY RETRIEVAL		COMPETING KEYS RETRIEVAL	
SWA NoPE	65.7	RoPE	91.8
SWA RoPE	37.9	SWA RoPE	75.6
RoPE	30.5	SWA NoPE	67.6

Reversal is aligned with mechanism shift

Open-Weight Model Sweep

Model	PE Configuration	Window Size	Local:Global
RoPE Architectures			
Gemma-2 (Riviere et al., 2024)	RoPE (10K)	4096	1:1
Llama-3.1 (Grattafiori et al., 2024)	RoPE (500K)	—	—
Qwen-2.5 (Yang et al., 2024)	RoPE (1M)	—	—
Qwen-3 (Yang et al., 2025a)	RoPE (1M)	—	—
PE Hybrid Architectures			
Gemma-3 (Kamath et al., 2025)	RoPE (10K) / RoPE (1M)	1024	5:1
Gemma-4 (Gemma, 2026) [†]	RoPE (10K) / p-RoPE (1M)	512 1024	4:1 5:1
SmolLM3 (Bakouch et al., 2025)	RoPE (5M) / NoPE	—	3:1
Command R7B (Yang et al., 2025b)	RoPE (50K) / NoPE	4096	3:1

PE Hybrids Are Consistently More Semantic



What does this tell us?

- PE choice shapes the mechanisms models use for in-context retrieval
- These underlying mechanisms shape long-context retrieval performance
- Actionable Interpretability: Understanding model internals enables intentional design

Open Questions

- How does the data distribution shape mechanism allocation?
- How do linear hybrids shape retrieval mechanisms?